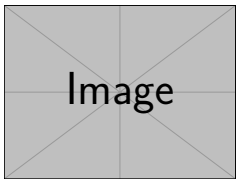
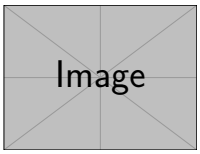
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A [Self-Adjusting / Adaptive] Decision Support System for Strategic [Domain] Optimization Using Integrated [Method A] and [Method B]

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ABSTRACT

This study proposes a novel [Framework Name] that integrates [Method A] and [Method B]. The system is designed to optimize [problem domain] under dynamic and uncertain conditions. The framework addresses a critical challenge in strategic decision-making by combining [static component] with a [dynamic component] that adaptively adjusts its actions in response to real-time environmental feedback. [Method A]-derived weights for Key Performance Indicators (KPIs) are incorporated into a strategically designed reward function that guides the [agent/algorithm], operating under an ε-greedy policy, to learn an optimal long-term [allocation/selection] strategy. This ensures that adaptive decisions remain aligned with organizational priorities while responding flexibly to operational changes. The framework was validated via [simulation platform], demonstrating a [X]% improvement in cumulative [metric] compared to conventional non-adaptive approaches. Results indicate that the [system] consistently generates high-quality, data-driven decisions, enhancing [KPI 1], [KPI 2], and long-term [outcome]. This framework provides actionable insights for managers and decision-makers seeking to integrate strategic goal-setting with adaptive operational decision-making in complex and uncertain environments.

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1. Introduction

[Topic area] is a core organizational capability, essential for aligning investments with long-term strategic goals and achieving organizational excellence (Author, 2021; Author et al., 2016). In today's dynamic and uncertain business environment, organizational success increasingly depends on effective and timely strategic decision-making, often supported by robust Decision Support Systems (DSS) (Author & Carvalho, 2024; **?**). This requires continuous development of DSS and [domain] functionalities to maintain organizational agility and respond effectively to environmental changes (Author et al., 2023; **?**; **?**).

While structured [domain] methodologies play a vital role in establishing evaluation criteria and processes (**??**), their effectiveness is fundamentally challenged by environmental volatility (**??**). Relying on static multi-criteria decision-making (MCDM) models (e.g., simple [Method A] or [Variant]) leads to strategic rigidity in resource allocation, as the strategic weights of criteria fail to update quickly enough with shifting operational conditions and emerging risks (**?**). This situation necessitates a critical evaluation of decision support architectures within [context] firms. Consequently, a key management challenge arises: How can an automated and rational Adaptive Decision Support System (ADSS)

be designed to dynamically adjust [domain] criteria in real-time, ensuring continuous alignment with evolving environmental conditions and strategic priorities?

Significant advancements in intelligent methods, particularly Reinforcement Learning (RL), offer a powerful new paradigm for sequential and dynamic decision problems (**?**). RL has demonstrated strong performance and adaptability in optimizing dynamic resource allocation in complex systems as well as in financial portfolio management (Author et al., 2025), thereby boosting organizational resilience. The development of hybrid DSS architectures that integrate structured approaches like MCDM with optimization techniques is vital for effective project selection in complex environments (**?**). Nevertheless, the [domain] literature lacks a coherent hybrid ADSS model that successfully integrates formal strategic preference setting (i.e., managerial KPI weighting) with dynamic operational decision-making via RL.

This research addresses this critical gap by introducing a novel [Method B]–[Method A] hybrid framework for Adaptive Decision Support Systems (ADSS). The core contribution lies in synergistically integrating the [Algorithm] (for dynamic, autonomous action selection) with the [Method A]. [Method A] serves as a robust tool to calculate validated KPI weights, ensuring the RL system's reward function is directly and authentically tied to management's strategic priorities (**?**). This combination not only enhances decision quality in the face of uncertainty but also provides a near-optimal policy for [domain] management that consistently remains aligned with the organization's long-term strategic direction.

Results from a dynamic simulation using [Platform] demonstrate that the [Method B]–[Method A] hybrid framework significantly enhances strategic alignment and [metric] compared to static methods, providing [domain] managers with a self-adjusting DSS for effective decision-making in uncertain environments.

2. Literature Review

2.1. Theoretical Foundations and Requirements for Adaptive Decision Support Systems (ADSS)

In the management literature, [Domain] Management is increasingly viewed not merely as a mechanical tool for [selection] but as a Governance Structure and a Strategy Enactment mechanism. Effective [domain] management enables organizations to achieve Ambidexterity by managing the simultaneous pursuit of exploration and exploitation (**?**). This strategic requirement directly translates into the necessity for Adaptive Decision Support System (ADSS) architectures that can computationally support dynamic strategy.

Portfolio Governance plays a key role in establishing transparency, allocating authority, and creating clear processes for strategic decision-making (**?**). From a systems perspective, Dynamic Capabilities Theory provides the necessary theoretical justification for strategic flexibility in [domain], emphasizing the organization's ability to sense, seize, and transform resources to maintain agility (**?**). These capabilities are vital for organizational excellence and sustained survival in fast-paced, competitive environments characterized by technological convergence and market uncertainty (**?**). The central challenge for DSS design is developing a computational framework that can accurately and autonomously operationalize these dynamic capabilities.

2.2. The Limitations of Multi-Criteria Decision-Making (MCDM) Models in Dynamic Decision Support

Traditional decision support for resource allocation in [domain] is primarily guided by Multi-Criteria Decision-Making (MCDM) models. The Analytic Hierarchy Process ([Method A]), founded by Saaty (1990), is recognized as a powerful tool for converting subjective qualitative managerial judgments into quantitative relative weights for decision criteria (**?**). The key value of [Method A] and its family (e.g., ANP) lies in their ability to: (1) manage consistency in managerial judgments and (2) provide transparency in the prioritization process.

However, the main structural weakness of [Method A] and its derivatives is their inherent inability to manage the sequential and adaptive nature of [domain] decisions (**?**). [Domain] decisions span long time horizons and necessitate frequent re-evaluation (**?**). Static MCDM methods offer only a "snapshot" of strategic preferences at a single point in time, failing to internalize the dynamic changes in criteria weights and resource constraints throughout the [domain] lifecycle. This deficiency leads to a critical breakdown in decision support efficacy (**??**).

2.3. Reinforcement Learning (RL) as a Computational Solution for Sequential Decision Systems

To address the shortcomings of static decision support models, the scientific literature has shifted toward algorithmic frameworks built upon Markov Decision Processes (MDPs) (**?**). Reinforcement Learning (RL), rooted in control theory and machine learning, is the most robust and suitable algorithmic paradigm for solving sequential decision problems in dynamic environments (Sutton & Barto, 1998; **?**).

RL, especially the Q-Learning algorithm developed by Watkins & Dayan (1992), allows the intelligent agent to learn an optimal policy without prior knowledge of the environment model (model-free learning) through repeated interaction and receiving a Reward signal (**?**). RL applications in related management fields are growing rapidly and prove its efficacy in adaptive decision-making:

- **Resource Management:** RL is used for optimizing dynamic resource allocation in complex systems, outperforming conventional methods (**?**).
- **Operations & Supply Chain:** In dynamic inventory and supply chain management, RL enables real-time, adaptive control policies (Author, 2022).
- **Financial Portfolio:** RL has demonstrated strong performance in selecting financial portfolios under high market volatility and sequential risk.

Despite these advancements, a central gap persists in [domain] research and the design of hybrid DSS architectures: existing RL implementations in management simplify the problem by assuming pre-defined, typically financial, reward functions (**?**). Without a structured mechanism to tie high-level strategic preferences ([Method A] output) to the dynamic reward signal (RL input), the learning system may drift, leading to policies that are operationally efficient but strategically misaligned (Author, 2023). This constitutes a significant system design challenge for adaptive DSS.

2.4. Synthesis and Research Contribution

This research directly addresses this theoretical and practical gap by formally integrating the prescriptive power of [Method A] (representing static strategic thought) and the adaptive capability of RL (representing dynamic operational decision-making) (**?**).

The main contribution of this study is the creation of a novel [Method B]–[Method A] hybrid Adaptive Decision Support System (ADSS) framework where the [Method A] output (strategic criteria weights) serves as a Strategic Filter in designing the Q-Learning reward function. This dual mechanism provides the first system that: (1) possesses the agility required to respond to environmental changes (enhancing dynamic capabilities) and (2) ensures that all dynamic [domain] actions are consistently aligned with the approved strategic vision and goals of senior management (**?**).

3. Methodology: [Framework Name] (ADSS) Framework

This section presents the methodological design of the proposed [Method B] – [Method A] Adaptive Decision Support System (ADSS).

3.1. Overview and Formalization of the Hybrid ADSS Architecture

The proposed model is a [Method B]–[Method A] hybrid Adaptive Decision Support System (ADSS) framework designed to optimize [domain] selection under uncertainty dynamically. The architecture formally integrates the prescriptive power of Multi-Criteria Decision-Making (MCDM) with the adaptive capability of Reinforcement Learning (RL) to address the limitations of static decision-making in turbulent environments. To directly resolve this limitation, the proposed [Method B]–[Method A] ADSS incorporates a Strategic Filter that evaluates and constrains each candidate RL action against the strategic hierarchy defined through [Method A]. The sequential decision problem is modeled as a Markov Decision Process (MDP), defined by the tuple $\langle S, A, T, R \rangle$.

3.2. Markov Decision Process (MDP) Formulation

Q-learning was selected as the learning algorithm due to the fully discrete and relatively low-dimensional state–action space of the [domain] environment, which makes tabular methods both feasible and efficient. Unlike DQN-based approaches, which are designed for high-dimensional or continuous state spaces requiring function approximation, the present setting does not necessitate neural network estimation. Q-learning also offers an advantage over SARSA in this application because its off-policy nature enables stable learning of a target greedy policy while allowing stochastic exploratory behavior during training. Furthermore, the tabular Q-learning framework provides full transparency and interpretability of learned values, which is essential for managerial decision-support contexts.

3.2.1. Q-Learning Algorithm. In this study, we employ Reinforcement Learning (RL) to enable adaptive decision-making in [domain] management under uncertainty. RL is well-suited for dynamic environments, allowing agents to learn through trial and error by interacting with the environment and adjusting their policy accordingly (Sutton & Barto, 1998).

Specifically, we utilize the Q-learning algorithm, a model-free method where the agent observes the current state (defined by performance indicators), selects an action, receives a reward, and iteratively updates its Q-values. The Q-Learning algorithm determines the optimal state-action mapping through gradual learning. Continuous adaptation is achieved via the Q-value update rule:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_{a'} Q(S_{t+1}, a') - Q(s_t, a_t) \right] \tag{1}$$

where:

- α is the learning rate $(0 < \alpha < 1)$, determining how quickly the algorithm adapts to new data,
- γ is the discount factor $(0 < \gamma < 1)$ for future rewards,
- R_t is the reward received at time step t,
- $Q(s_t, a_t)$ is the current value of the state-action pair (s_t, a_t), and $\max Q(S_{t+1}, a')$ represents the maximum Q-value over all possible actions a' in the next state S_{t+1}.

Algorithm 1 Q-Learning Approach

Input: State space S, action space A, learning rate α, discount factor γ, exploration rate ε
Output: Optimized Q-value function $Q(s,a)$

Initialize Q-values: $Q(s,a) \leftarrow 0, \forall s \in S,\ a \in A$
repeat
  Observe current state $s = s_t$
  Select action a_t using ε-greedy policy:
    With probability ε, choose a random action (exploration)
    With probability $1-\varepsilon$, choose $a_t = \arg\max_a Q(s,a)$ (exploitation)
  Execute action a_t, observe reward r_t, and next state $s' = s_{t+1}$
  Update Q-value using Equation (1)
  Set $s \leftarrow s'$
until convergence criterion is met (e.g., change in Q-values $<$ threshold)

The State Space is defined by three discrete and mutually exclusive states: $S = \{\text{Normal}, \text{Crisis}, \text{Sudden Change}\}$. These states collectively capture the overall [domain] health and the underlying environmental uncertainty context. The Action Space includes N discrete, [domain]-level strategic policies representing management's responses to environmental states.

3.3. Hybrid Reward Function Design and Strategic Aggregation

Unlike conventional RL frameworks with predefined reward functions, this study introduces a hybrid reward formulation directly grounded in managerial strategic weights derived from [Method A]. The framework's core contribution is the Hybrid Reward Function (R_{total}), which directly aligns the RL agent's learning with management's strategic preferences ([Method A] weights). Its calculation involves two stages: Immediate KPI Reward Estimation and Strategic Aggregation.

3.3.1. Immediate KPI Reward Estimation. The observed reward for a KPI, $r_{n,i}$, is based on its percentage deviation from the strategic target. To ensure the reward reflects stable strategic changes rather than short-term fluctuations, a Smoothed Strategic Performance Score (R_n) is calculated using a rolling window k:

$$R_n = \frac{1}{k}\sum_{i=1}^{k} r_{n,i} \tag{2}$$

The corresponding reward structure based on action outcome is summarized in Table 2.

3.3.2. Strategic Aggregation and Temporal Weighting. The final reward received by the agent at each time step is the Total Portfolio Reward (R_{total}). It is derived from the strategically weighted aggregation of the stabilized performance scores (R_n) using the [Method A]-derived strategic weights (w_n):

$$R_{\text{total}} = \sum_{n=1}^{K} w_n R_n = w_1 R_1 + w_2 R_2 + \cdots + w_K R_K \tag{3}$$

3.4. Implementation Steps of the Hybrid Framework

The proposed decision-making framework integrates the static, prescriptive power of [Method A] with the adaptive capability of the Q-Learning algorithm. The implementation is executed in two distinct phases:

Phase 1: Strategic Pre-Processing ([Method A]). (1) Identification and Weighting of KPIs: Define the set of the n-th Key Performance Indicator (KPI_n) that represents the organizational strategic goals. The relative importance (strategic weight, w_n) is formally computed through [Method A], ensuring managerial consensus and consistency. (2) State and Action Definition: Analyze [domain] performance thresholds to define the environmental states and the corresponding [domain] actions available to the agent.

Phase 2: Iterative Learning (Q-Learning). (1) Initialization: Initialize the Q-Table $Q(s,a) \leftarrow 0$ and set the learning parameters $(\alpha, \gamma, \varepsilon)$. (2) Reward Computation: During each iteration, compute the estimated expected reward (R_n) for each KPI using Equation (2). (3) Hybrid Reward Aggregation: Aggregate into R_{total} via Equation (3). (4) Optimal Action Selection: Select a_t based on s_t using the ε-greedy policy. (5) Q-Value Update: Execute, observe S_{t+1}, and update using Equation (1). (6) Policy Convergence: Repeat until convergence.

3.5. Case Study: Scenario-Based [Domain] Evaluation

To validate the hybrid model, a scenario-based simulation was conducted at a major [type] organization. Three representative environmental shock scenarios were simulated. Shock 1 represents a [Priority] Shock, modeled by increasing the [KPI] weight by 40% while proportionally reducing other strategic weights. Shock 2 represents an Operational Crisis Shock, modeled by reducing [KPI A] and [KPI B] reward contributions by approximately 30%. Shock 3 represents a [Regulatory/Sustainability] Shock, modeled by increasing the corresponding KPI weights by approximately 35%.

Nine performance indicators (KPIs) were formulated:

1. **KPI 1:** [Definition]
2. **KPI 2:** [Definition]
3. **KPI 3:** [Definition]
4. **KPI 4:** [Definition]
5. **KPI 5:** [Definition]
6. **KPI 6:** [Definition]

7. **KPI 7:** [Definition]
8. **KPI 8:** [Definition]
9. **KPI 9:** [Definition]

4. Results and Discussion

4.1. Analytical Hierarchy Process ([Method A]) for KPI Weighting

To quantify the relative strategic importance of each KPI, a structured pairwise comparison method was applied using Saaty's 9-point scale. The evaluations were obtained from five senior [domain] managers and strategy experts. Following standard [Method A] computation, the matrix was normalized, and the mean of each row was calculated to derive the priority vector representing the relative weights of the nine KPIs. The resulting weights (Table 4) indicate that [KPI 1] (0.XXXX) and [KPI 2] (0.XXXX) dominate the strategic preference structure.

To verify the internal consistency of expert judgments, the Consistency Index (CI) was computed:

$$CI = \frac{\lambda_{\max} - n}{n-1} \tag{4}$$

The Consistency Ratio (CR) was then obtained by dividing CI by the Random Index (RI). According to Saaty's scale, the RI for $n = 9$ is 1.45:

$$CR = \frac{CI}{RI} \tag{5}$$

A matrix is considered consistent if $CR < 0.1$. In this study, the calculated CR satisfied this condition, confirming that the expert evaluations were logically consistent.

4.2. State–Action Pair Matrices for Key Indicators

The state–action matrices for key performance indicators align with the organization's long-term goals. Proposed actions enhance resilience to shocks, strengthen [domain] systems, and improve service quality, balancing short- and long-term strategy.

4.3. Q-Learning Algorithm Implementation and Policy Convergence

4.3.1. Parameter Configuration and Update Rule. To ensure convergence stability and emphasize long-term strategic outcomes, the learning rate (α) was set to 0.1, and the discount factor (γ) was set to 0.9. The ε-greedy policy was implemented with an initial value of 1.0, linearly decaying to 0.01 over the training episodes to ensure a robust balance between exploration and exploitation.

4.3.2. Policy Convergence and Performance Interpretation. Convergence behavior was evaluated through the stabilization of Q-values and the diminishing variance of cumulative rewards over successive training episodes. The convergence patterns demonstrated smooth and monotonic learning trajectories across all KPIs, indicating robust policy stabilization and the absence of oscillatory behavior. Stabilization of the total reward after approximately 400 episodes confirms that the agent successfully learned an optimal, steady-state strategic policy.

4.4. Strategic Reward Aggregation and Final Reward Calculation

To calculate the total weighted reward, we apply Equation (3), in which each indicator's average reward is multiplied by its corresponding relative weight:

$$R_{\text{total}} = (0.3521 \times 1.556) + (0.1951 \times 2.0) + \cdots + (0.0272 \times 1.889) = 1.73 \tag{6}$$

4.5. Ex-Post Analysis of KPI Performance Contribution

Although the strategic priorities (W_n) are fixed and exogenously determined by [Method A], it is valuable to conduct an ex-post analysis to evaluate the relative contribution of each KPI's performance (R_n) to the overall reward structure. The relative operational contribution (C_n) of each indicator is calculated as:

$$C_n = \frac{R_n}{\sum_{i=1}^{9} R_i} \tag{7}$$

The analysis shows [KPI X] contributed most to the total average reward (Table 5), followed by [KPI Y], indicating that the RL agent effectively managed these areas.

4.6. Policy Convergence and Sensitivity Analysis

This section presents simulation results from the proposed reinforcement learning model. Figure 1 illustrates the exploitation-only policy ($\varepsilon = 0$) maintaining higher cumulative reward values compared to the exploratory policy ($\varepsilon = 0.3$) throughout the simulation. At the final episode (Episode 200), the exploitation strategy reaches a cumulative reward of approximately 950, whereas the exploratory policy reaches approximately 780. This ≈21% difference reflects the expected "exploration cost."

Figure 3 presents the distribution of R_{total} over 50 independent runs. The boxplot confirms that the Hybrid [Method B]–[Method A] policy has the highest median, the tightest spread, and the fewest extreme deviations, supporting its robustness and repeatability.

4.7. Comparative Performance Analysis

The proposed hybrid [Method B]–[Method A] policy was benchmarked against three non-adaptive control scenarios:

1. **[Method A]-Only Policy (Control 1):** A static decision-making strategy based solely on [Method A]-derived priorities, without the ability to adapt to changing states.
2. **Random Policy (Control 2):** A non-intelligent baseline where actions are selected randomly in every state.
3. **Simple Weighted-Sum Heuristic (Control 3):** A classical multi-criteria scoring method in which each candidate action is evaluated using a deterministic weighted-sum score.

The comparative performance metrics are presented in Table 6. The results clearly demonstrate:

1. **Superior Performance:** The model achieves the highest strategic performance ($R_{\text{total}} = 1.7359$), confirming that the adaptive policy outperforms non-adaptive alternatives.
2. **Gain over Static Policy:** The hybrid model improves the strategic weighted reward compared to the static-only policy (1.5500) by selecting optimal corrective actions during crisis or sudden change states.
3. **Baseline Validation:** The random policy ($R_{\text{total}} = 0.8500$) performs poorly, confirming that the learning mechanism and reward design effectively guide the agent toward high-performance outcomes.

4.8. Statistical Validation and Robustness Analysis

To strengthen the validation, two additional analyses were conducted: multi-run stability testing and environmental shock robustness testing. The model was executed 50 independent times, each for 200 episodes, under the same parameter setting ($\alpha = 0.1$, $\gamma = 0.9$). The aggregated statistical results across the three policies are presented in Table 7. The Hybrid [Method B]–[Method A] model achieves the highest mean reward while also exhibiting the lowest variability.

4.9. Limitations and Computational Considerations

While the proposed [Method B]–[Method A] framework is computationally tractable for the [domain] size and KPI structure examined in this study, the methodology may face scalability challenges in substantially larger [domain] environments. As the number of assets, KPIs, and admissible reallocation options expands, the dimensionality of the state–action space grows non-linearly. Future work may address these limitations through state aggregation, function approximation, or migrating toward deep reinforcement learning variants (e.g., DQN) to enhance scalability and computational efficiency.

5. Conclusion

This study developed and validated a hybrid decision-making framework integrating [Method A] and Q-Learning to optimize [domain] management under uncertainty. Its core contribution is the Strategic Hybrid Reward Function, which maps [Method A]-based strategic priorities onto the agent's learning objectives, ensuring adaptive and aligned policies.

Empirical validation demonstrated that the Q-values converged steadily and the cumulative weighted rewards stabilized, confirming the model's effectiveness in learning optimal policies focused on high-priority indicators, particularly [KPI X]. Sensitivity analyses on key parameters (α, γ, ε) highlighted the importance of careful tuning for stable convergence and efficient long-term learning.

Comparative performance analysis showed that the hybrid model achieved a higher Total Weighted Reward (R_{total}) than static-only policies, demonstrating its ability to learn adaptive, superior strategies that yield greater strategic returns in dynamic environments.

Future research should focus on:

- Extending the framework toward deep reinforcement learning approaches capable of handling larger and more complex [domain] environments, including richer and higher-resolution state representations.
- Applying the model to real organizational datasets once such access becomes available, enabling empirical validation under real-world [domain] management conditions.
- Extending the benchmarking analysis by comparing the proposed hybrid [Method B]–[Method A] framework with advanced established [domain] optimization techniques (e.g., mean–variance optimization, linear or integer programming models).

Table 1. *Sample Q-Table.*

States	A_1	A_2	$\cdots$	A_m
S_1	Q_{11}	Q_{12}	$\cdots$	Q_{1m}
S_2	Q_{21}	Q_{22}	$\cdots$	Q_{2m}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
S_n	Q_{n1}	Q_{n2}	$\cdots$	Q_{nm}

Table 2. *Reward assignment based on action outcomes.*

Reward	Action Outcome
+6	More than 20% positive deviation from the target
+5	15% to 20% positive deviation
+4	10% to 15% positive deviation
+3	5% to 10% positive deviation
+2	Up to 5% positive deviation
+1	No deviation from the target
0	Very slight shortfall (0% to 1% negative deviation)
−1	Up to 5% negative deviation
−2	5% to 10% negative deviation
−3	10% to 15% negative deviation
−4	15% to 20% negative deviation
−5	More than 20% negative deviation

Table 4. *Criteria weights.*

Weight	Indicator
0.3521	KPI 1 ([Name])
0.1951	KPI 2 ([Name])
0.1047	KPI 3 ([Name])
0.0989	KPI 4 ([Name])
0.0768	KPI 5 ([Name])
0.0751	KPI 6 ([Name])
0.0475	KPI 7 ([Name])
0.0272	KPI 8 ([Name])
0.0225	KPI 9 ([Name])

Table 5. *Relative operational contribution of indicators.*

Indicator	R_n	C_n
KPI 1	1.556	0.097
KPI 2	2.000	0.125
KPI 3	1.667	0.104
KPI 4	1.222	0.076
KPI 5	2.111	0.132
KPI 6	2.333	0.146
KPI 7	1.778	0.111
KPI 8	1.333	0.083
KPI 9	1.889	0.118

Table 7. *Multi-run statistical validation of the three decision policies.*

Scenario	Mean	SD	Var	CV	95% CI
Hybrid [B]–[A]	1.732	0.041	0.00168	0.024	[1.720, 1.744]
[A]-Only	1.548	0.079	0.00624	0.051	[1.526, 1.570]
Random	0.861	0.133	0.01769	0.155	[0.824, 0.898]

Table 8. *Robustness results under environmental shocks.*

Scenario	Mean R_{total}	Change vs. Baseline	SD
Hybrid [Method B]–[Method A] (no shock)	1.732	—	0.041
Shock 1: [Priority] Shock	1.688	−2.54%	0.045
Shock 2: Operational Crisis Shock	1.642	−5.20%	0.049
Shock 3: [Regulatory] Shock	1.703	−1.68%	0.043

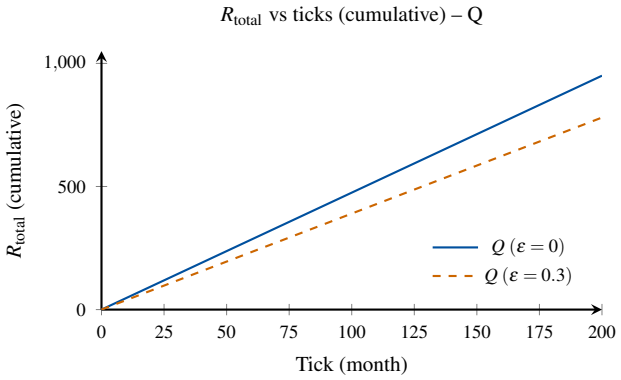


Figure 1. *Comparison of R_{total} cumulative values for $\varepsilon = 0$ (exploitation) vs. $\varepsilon = 0.3$ (exploration) over the simulation horizon.*

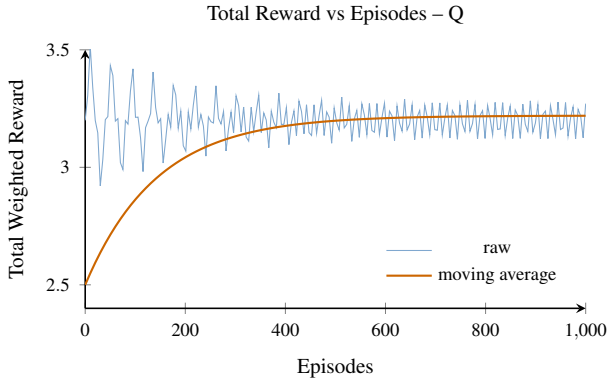


Figure 2. *Total reward per episode over time.*

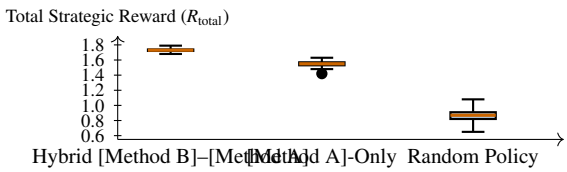


Figure 3. *Boxplot of R_{total} over 50 runs for the three policies.*

Author contributions

Table 3. *Pairwise comparison matrix.*

	KPI 1	KPI 2	KPI 3	KPI 4	KPI 5	KPI 6	KPI 7	KPI 8	KPI 9
KPI 1	1	3	5	7	9	4	6	5	8
KPI 2	1/3	1	3	5	7	2	4	3	6
KPI 3	1/5	1/3	1	3	5	1	2	2	4
KPI 4	1/7	1/5	1/3	1	3	1/2	1	1/3	2
KPI 5	1/9	1/7	1/5	1/3	1	1/4	1/5	1/6	1
KPI 6	1/4	1/2	1	2	4	1	2	2	3
KPI 7	1/6	1/4	1/2	1	5	1/2	1	2	4
KPI 8	1/5	1/3	1/2	3	6	1/2	1/2	1	3
KPI 9	1/8	1/6	1/4	1/2	1	1/3	1/4	1/3	1

Table 6. *Comparative performance metrics for [domain] management scenarios.*

Model Scenario	Strategy Description	R_{total}	Gain vs. [A]-Only
Hybrid [B]–[A]	Optimal, adaptive policy learned through Q-Learning ($\alpha = 0.1$, $\gamma = 0.9$)	1.7359	+11.99%
Weighted-Sum Heuristic (Ctrl 3)	Classical scoring-based static optimisation using [A] weights	1.6120	+4.00%
[A]-Only Policy (Ctrl 1)	Static, non-adaptive policy (Simulated Result)	1.5500	—
Random Policy (Ctrl 2)	Actions selected randomly in each state (Simulated Baseline)	0.8500	—

CRediT: **First Author:** Data curation, Methodology, Validation, Visualization, Writing – original draft; **Second Author:** Investigation, Methodology, Project administration, Supervision, Validation, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data supporting the findings of this study were generated through simulation experiments. The simulation code and related materials are openly available in Zenodo at https://doi.org/10.5281/zenodo.XXXXXXX.

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